An explicit gap for three-point weighted selection in planar vector-valued regression

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10 October 2026

Abstract

We prove $F_{\text{weighted}}(2,2,3) \le 11/6 - 10^{-20}$ for minimum-Frobenius-norm least-squares regression with two input and two output coordinates. The argument combines a scalar certificate with quantitative stability of centered planar selection and a seven-atom support reduction. Together with the known lower bound, this gives $13/8 \le F_{\text{weighted}}(2,2,3) \le 11/6 - 10^{-20}$. Lean verifies the scalar selection theorem, its anisotropy improvement, and exact numerical implications; the geometric argument is proved here.

1 Problem and result

Weighted data selection for regression was proposed by Hanneke, Moran, Shlimovich and Yehudayoff [1, 2]. Zhang [3], Theorem 6.4, established $13/8 \le F_{\text{weighted}}(2,2,3) \le 15/8$ and reduced the conjectured value $13/8$ to a finite planar moment problem. We improve the upper bound.

For a finite dataset $D = ((x_i, y_i))_{i=1}^N \subset \mathbb{R}^2 \times \mathbb{R}^2$, let $L_D(W) = N^{-1} \sum_i \|Wx_i - y_i\|^2$ and $L_D^* = \min_W L_D(W)$. For nonnegative weights c of sum one, let $\widehat{W}_c$ be the minimum-Frobenius-norm minimizer of $\sum_i c_i \|Wx_i - y_i\|^2$. Then $F_{\text{weighted}}(2,2,3)$ is the supremum over datasets of

$$\frac{\inf_{|\operatorname{supp} c| \le 3} L_D(\widehat{W}_c)}{L_D^*},$$

with $0/0 = 1$. The infimum, including limits of full-rank selections, is essential.

A residual frame is a finite collection $p_i > 0$, $u_i, w_i \in \mathbb{R}^2$ with

$$\sum_i p_i = 1, \quad \|u_i\|^2 = 2, \quad \sum_i p_i u_i u_i^\top = I_2, \quad \sum_i p_i w_i u_i^\top = 0. \tag{1}$$

Put $E = \sum_i p_i \|w_i\|^2$ and $R = \sum_i p_i w_i w_i^\top$. For independent u_i, u_j, let $G_{ij} u_i = w_i$, $G_{ij} u_j = w_j$. Using Frobenius distance, define

$$\mathcal{T}_3 = \min_{\substack{2 \le |S| \le 3 \\ S \text{ contains an independent pair}}} \operatorname{dist}\left(0, \operatorname{conv}\{G_{ij} : i, j \in S,\ \det(u_i, u_j) \ne 0\}\right)^2.$$

Tightness guarantees an independent pair.

Theorem 1. *Every residual frame satisfies $\mathcal{T}_3 \le (5/6 - 10^{-20})E$. Consequently $F_{\text{weighted}}(2,2,3) \le 11/6 - 10^{-20}$.*

2 Planar selection and the scalar certificate

Lemma 2. *Let $v_j \in \mathbb{R}^2$ have positive probabilities p_j and mean zero. For any real 2×2 matrix A with $\operatorname{tr} A = 1$, some convex combination v of at most two of these points satisfies*

$$\|v\|^2 \le \sum_j p_j \|Av_j\|^2.$$

Proof. Minimize the right-hand linear objective over centered probability vectors on the fixed points, and choose a minimizer q of least support. If the columns $(1, v_j)$ on its positive support are dependent, a supported kernel vector has both signs. Small perturbations in either direction remain feasible, so optimality forces zero objective slope. Continuing until one weight vanishes contradicts least support. Hence at most three affinely independent points remain and their objective is no larger than the original. If at most two remain, their centered mean itself is zero.

Otherwise let $S_q = \sum_j q_j v_j v_j^\top$. It is positive definite: a null direction would place all three points on a line. Put $n_j = S_q^{-1} v_j$, choose i maximizing $r = \|n_i\|^2 > 0$, and set

$$\lambda_j = q_j \left(1 - \frac{n_j^\top n_i}{r}\right).$$

Cauchy–Schwarz and maximality give $\lambda_j \geq 0$. Centering gives $\sum_j \lambda_j = 1$, and $\lambda_i = 0$. Since $\sum_j q_j v_j n_j^\top = I_2$,

$$v = \sum_j \lambda_j v_j = -\frac{n_i}{r}, \qquad \|v\|^2 = \frac{1}{r}.$$

Finally, writing $C_q = \sum_j q_j \|A v_j\|^2$,

$$1 = \sum_j q_j n_j^\top A v_j, \qquad 1 \leq \left(\sum_j q_j \|n_j\|^2\right) C_q \leq r C_q.$$

Thus $\|v\|^2 \leq C_q$, proving the assertion. □

For positive semidefinite S with $\operatorname{tr} S > 0$, the matrix $A = \operatorname{adj} S / \operatorname{tr} S$ has trace one, and in dimension two

$$\operatorname{tr}(ASA^\top) = \frac{\det S}{\operatorname{tr} S}. \tag{2}$$

This identity holds also when S is singular. If $\operatorname{tr} S = 0$, all positive-weight points vanish. Thus singular covariances cause no gap.

Fix an anchor i and a unit vector $n_i \perp u_i$. Put

$$M_i = \tfrac{1}{2} w_i u_i^\top, \quad b_{ij} = u_j^\top n_i, \quad c_{ij} = \tfrac{1}{2} u_i^\top u_j, \quad \xi_{ij} = \frac{w_j - c_{ij} w_i}{b_{ij}} \quad (b_{ij} \neq 0).$$

Because $u_j = c_{ij} u_i + b_{ij} n_i$,

$$G_{ij} = M_i + \xi_{ij} n_i^\top, \qquad \|M_i + \xi n_i^\top\|_F^2 = \tfrac{1}{2}\|w_i\|^2 + \|\xi\|^2.$$

The cross term vanishes since $u_i^\top n_i = 0$. The positive star weights $P_{ij} = p_j b_{ij}^2$ satisfy

$$\sum_{b_{ij} \neq 0} P_{ij} = 1, \qquad \sum_{b_{ij} \neq 0} P_{ij} \xi_{ij} = 0.$$

Indeed these are respectively $n_i^\top I_2 n_i = 1$ and the difference between the cross-moment $\sum_j p_j w_j u_j^\top n_i = 0$ and $w_i u_i^\top I_2 n_i / 2 = 0$.

Define

$$L_i = \sum_{b_{ij} = 0} p_j (w_j - c_{ij} w_i)(w_j - c_{ij} w_i)^\top \succeq 0, \qquad H_i = R + \tfrac{1}{2} w_i w_i^\top.$$

The star covariance is

$$S_i = \sum_{b_{ij} \neq 0} P_{ij} \xi_{ij} \xi_{ij}^\top = H_i - L_i. \tag{3}$$

To check it, expand the same sum over all j. The two cross terms vanish because $\sum_j p_j c_{ij} w_j = 0$, and $\sum_j p_j c_{ij}^2 = 1/2$. Subtracting the parallel terms yields (3).

Apply Lemma 2 with $A_i = \operatorname{adj} H_i / \operatorname{tr} H_i$; its denominator $E + \|w_i\|^2/2$ is positive for $E > 0$. PSD monotonicity and (2) give a two-neighbor convex mean with

$$\|\xi\|^2 \leq \operatorname{tr}(A_i S_i A_i^\top) = \frac{\det H_i}{\operatorname{tr} H_i} - \operatorname{tr}(A_i L_i A_i^\top).$$

Consequently, for every anchor,

$$\mathcal{T}_3 \le \frac{\|w_i\|^2}{2} + \frac{\det(R + w_i w_i^\top/2)}{E + \|w_i\|^2/2} - \operatorname{tr}(A_i L_i A_i^\top). \tag{4}$$

This involves at most three atoms. Dropping the last, nonnegative term is valid but potentially loses information.

Normalize $E = 1$ and rotate output coordinates so $R = \operatorname{diag}(r, 1-r)$. Set $a_i = w_{i1}^2$, $b_i = w_{i2}^2$, and $s = a + b$. The first two terms of (4) are

$$g_r(a,b) = \frac{s}{2} + \frac{r(1-r) + ((1-r)a + rb)/2}{1 + s/2}, \qquad \sum_i p_i a_i = r, \quad \sum_i p_i b_i = 1 - r.$$

Put $\delta = 2r - 1$, $x = 1 + 3\delta/10$, $y = 1 - 3\delta/10$, $P = xy$, $Q = 7/3 - 21\delta^2/100$, and $h = P - ay - bx$. Since

$$\sum_i p_i h(a_i, b_i) = \frac{21}{100}\delta^2 \ge 0,$$

some positive-weight atom has $h \ge 0$. Define

$$F(a,b) = s^2 + \frac{4}{3}s - \delta(a-b) - \frac{7}{3} - \delta^2.$$

Exact expansion gives

$$PF = -\frac{121}{100}\delta^2 P - h(Q + ax + by) - ab(x-y)^2, \tag{5}$$

$$4(1 + s/2)(g_r(a,b) - 5/6) = F(a,b). \tag{6}$$

Here $x, y, P, Q > 0$ for $|\delta| \le 1$. At the selected atom all terms on the right of (5) are nonpositive. This proves $\mathcal{T}_3 \le 5/6$. Homogeneity gives $\mathcal{T}_3 \le 5E/6$ for general energy.

We also need its quantitative anisotropy consequence. At $h \ge 0$, $s \le P/\min(x,y) = \max(x,y) \le 13/10$. Thus (5)–(6) give

$$\mathcal{T}_3 \le \frac{5}{6} - \frac{11}{60}\delta^2. \tag{7}$$

The coefficient is $(121/100)/(4(1 + 13/20)) = 11/60$.

The scalar relaxation alone is sharp: the one-atom scalar distribution $r = a = b = 1/2$ has $g = 5/6$. It is not a valid tight frame. The strict improvement must therefore use geometry.

3 Support reduction and quantitative stability

Lemma 3. *For each positive-energy frame there is a residual frame on a subset of at most seven of its fixed atoms, with energy $E' \le E$, and with $\mathcal{T}_3' \ge \mathcal{T}_3$. If $E' = 0$, then $\mathcal{T}_3 = 0$.*

Proof. Minimize $\sum_i q_i \|w_i\|^2$ over nonnegative weights on the fixed atoms satisfying the frame constraints. The symmetric input covariance gives three linear equations and the cross-covariance gives four. Normalization follows by taking the trace of the input equation, since $\|u_i\|^2 = 2$. Thus there are at most seven independent linear constraints. The feasible probability polytope is compact and nonempty. At a minimizing weight vector of least support, a dependence of its active constraint columns would permit both signs of small feasible perturbation. The energy slope must be zero by optimality. Continuing to the first zero weight would reduce support while preserving energy, a contradiction. The positive support consequently has at most seven atoms. Deleting atoms can only increase the selection minimum, giving $\mathcal{T}_3' \ge \mathcal{T}_3$. If $E' = 0$, all remaining w_i vanish; tightness still supplies an independent pair and its interpolator is zero. Then $\mathcal{T}_3' = 0$, so $\mathcal{T}_3 = 0$. □

Lemma 4. *For a finite centered planar probability distribution, let $\operatorname{tr} A = 1$, $C = \sum_j p_j \|A v_j\|^2 \le 1/2$, and let $m \ge 1/4$ be the minimum squared norm of a convex combination of at most two original points. Set $\zeta = C - m \ge 0$. For any original point v_* with $\|v_*\| \le 2$,*

$$C - \|A v_*\|^2 \le 50\zeta + 10\sqrt{\zeta} \le 60\sqrt{\zeta}.$$

Proof. Choose a centered least-support minimizer q of the linear A-cost. Since $m > 0$, its support is a positive triangle. Put $S_q = \sum_j q_j v_j v_j^\top$, $n_j = S_q^{-1} v_j$, $r = \max_j \|n_j\|^2$, $M = \sum_j q_j \|n_j\|^2$, $\alpha = 1/M$, and $C_q = \sum_j q_j \|Av_j\|^2$. The planar construction and weighted Cauchy–Schwarz yield

$$m \le 1/r \le \alpha \le C_q \le C.$$

Consequently $r \le 4$, $1/4 \le \alpha \le 1/2$, and

$$r - M \le 1/m - 1/\alpha \le \zeta/(m\alpha) \le 16\zeta.$$

For $e_j = Av_j - \alpha n_j$, the trace identity gives $\sum_j q_j \|e_j\|^2 = C_q - \alpha \le \zeta$. Write $c_j = \|Av_j\|^2$. Expanding and using Cauchy–Schwarz,

$$\sum_j q_j |c_j - \alpha^2 r| \le \alpha^2 (r - M) + 2\alpha\sqrt{r\zeta} + \zeta \le 5\zeta + 2\sqrt{\zeta},$$

$$|\alpha^2 r - C| \le \alpha^2 (r - M) + (C - \alpha) \le 5\zeta.$$

Thus $\sum_j q_j |c_j - C| \le 10\zeta + 2\sqrt{\zeta}$. The affine coordinates of v_* in the triangle are $\beta_j = q_j(1 + n_j^\top v_*)$: they sum to one and reproduce v_* by the covariance identities. Since $\|n_j\| \le 2$, $|\beta_j| \le 5q_j$. Inserting v_* with small positive weight while subtracting its affine coordinates from q is feasible. LP optimality implies $\|Av_*\|^2 \ge \sum_j \beta_j c_j$. Therefore

$$C - \|Av_*\|^2 \le \left| \sum_j \beta_j (c_j - C) \right| \le 50\zeta + 10\sqrt{\zeta}.$$

Finally $0 \le \zeta \le C \le 1/2 < 1$, so $\zeta \le \sqrt{\zeta}$. No bound on omitted points, or on triangle vertices themselves, was assumed. □

4 The explicit gap

Proof of the frame assertion in Theorem 1. If $E = 0$, all w_i vanish and $\mathcal{T}_3 = 0$. For $E > 0$, Lemma 3 and homogeneity reduce the proof to at most seven atoms with $E = 1$. Put $\epsilon = 10^{-20}$ and suppose $\mathcal{T}_3 \ge 5/6 - \epsilon$. By (7), $\delta^2 \le 60\epsilon/11$. Set $\eta = 2\epsilon + |\delta|$; then $\eta < 3/10^{10}$, since $60/11 < 25/4$.

For $s = \|w_i\|^2$ the anchor bound is at most $f(s) + |\delta|/2$, where

$$f(s) = s/2 + (1 + s)/(4 + 2s), \quad f(1) = 5/6, \quad 1/2 \le f'(s) = 1/2 + 1/(2(s + 2)^2) \le 5/8 \quad (s \ge 0).$$

Indeed $r(1 - r) \le 1/4$ and $((1 - r)a + rb)/2 \le (1 + |\delta|)s/4$. It follows that every $s_i \ge 1 - \eta$, and

$$\sum_i p_i (s_i - 1 + \eta) = \eta. \tag{8}$$

Hence if $p_i \ge 1/28$, then $1 - \eta \le s_i \le 1 + 27\eta$.

Retain these heavy atoms. The discarded mass is at most $1/4$. Writing $u_i/\sqrt{2} = (\cos\theta_i, \sin\theta_i)$, tightness says that $t_i = (\cos 2\theta_i, \sin 2\theta_i)$ has total mean zero. The retained conditional mean has norm at most $1/3$. There are therefore heavy i, j with $t_i^\top t_j \le -1/2$, or equivalently $c^2 = (u_i^\top u_j/2)^2 \le 1/4$. To see this, if zero lies in the retained hull, represent it using at most three circle points. An antipodal pair, or a cyclic angular gap in $[2\pi/3, \pi]$, gives the assertion. If zero is outside the hull, its nearest point lies on a boundary edge (a vertex has norm one). For unit endpoints the nearest point of their segment is its midpoint; the squared distance is $(1 + t_i^\top t_j)/2 \le 1/9$.

For the star at this heavy anchor, write $H = R + w_i w_i^\top/2$, $A = \operatorname{adj} H / \operatorname{tr} H$, $\kappa = \det H / \operatorname{tr} H$, $S = H - L$, $C = \operatorname{tr}(ASA^\top)$, and let m be the minimum two-point squared mean norm. Put $g_i = s_i/2 + \kappa$. Lemma 2 and the anchored certificates give $m \le C \le \kappa$ and $\mathcal{T}_3 \le s_i/2 + m$. Thus

$$0 \le \zeta = C - m \le \kappa - m \le g_i - \mathcal{T}_3 \le \frac{135}{8}\eta + |\delta|/2 + \epsilon \le 19\eta, \qquad m \ge 1/3 - \epsilon - 27\eta/2 \ge 1/4.$$

Put $v_k = w_k/\|w_k\|$ for $k = i, j$, and $H_0 = I_2/2 + v_i v_i^\top/2$, $A_0 = \operatorname{adj} H_0/(3/2)$. All subsequent estimates hold for $\eta \le 10^{-3}$, which is satisfied. Output diagonalization gives $\|R - I_2/2\|_{\mathrm{op}} = |\delta|/2$. Hence

$$\|H - H_0\|_{\mathrm{op}} \le 14\eta, \quad |\operatorname{tr} H - 3/2| \le 27\eta/2, \quad \operatorname{tr} H \ge 1.$$

H is positive definite, A has trace one, and $\kappa \le \operatorname{tr} H/4 \le 1/2$, so $C \le 1/2$. Writing $D = H - H_0$, the two-dimensional determinant expansion gives

$$|\det H - 1/2| \le (3/2)\|D\|_{\mathrm{op}} + \|D\|_{\mathrm{op}}^2 \le 22\eta.$$

It follows that $|\kappa - 1/3| \le 27\eta$. The identity $\operatorname{adj} H / \operatorname{tr} H = I_2 - H/\operatorname{tr} H$ gives $\|A - A_0\|_{\mathrm{op}} \le 23\eta$.

For the selected neighbor, put $b = b_{ij}$. Then

$$\xi = (w_j - c w_i)/b, \qquad \xi_0 = (v_j - c v_i)/b, \qquad b^2 = 2(1 - c^2) \ge 3/2.$$

The heavy norm bounds imply $\|\xi\| \le 2$, $\|\xi_0\| \le 3/2$, and $\|\xi - \xi_0\| \le 41\eta$; use $|\sqrt{s_k} - 1| \le |s_k - 1| \le 27\eta$ and $|c| \le 1/2$. As A is positive definite of trace one, its operator norm is at most one. Thus

$$\|A\xi - A_0\xi_0\| \le 87\eta, \qquad |\|A\xi\|^2 - \|A_0\xi_0\|^2| \le 305\eta.$$

For $d = v_i^\top v_j$, the eigenvalues of H_0^{-1} are $1, 2$, so

$$\|A_0\xi_0\|^2 = \frac{(d-c)^2 + 4(1-d^2)}{18(1-c^2)} \le \frac{26}{81}.$$

Indeed the numerator minus $(52/9)(1-c^2)$ equals $(64/9)c^2 - 16/9 - 3(d + c/3)^2 \le 0$. Consequently

$$\kappa - \|A\xi\|^2 \ge 1/81 - 332\eta \ge 1/162.$$

Here $\eta < 3/10^{10}$. Since $\kappa - C \le 19\eta$, we obtain

$$C - \|A\xi\|^2 \ge 1/162 - 19\eta \ge 1/200.$$

All hypotheses of Lemma 4 hold for this original star point. It forces

$$1/200 \le 60\sqrt{\zeta}, \qquad \zeta \ge 1/144000000.$$

But $\zeta \le 19\eta < 57/10^{10} < 1/144000000$, since $57 \cdot 144000000 = 8208000000 < 10^{10}$. This contradiction proves $\mathcal{T}_3 < 5/6 - 10^{-20}$ on the normalized reduced support, and the non-strict bound transfers to all finite frames by Lemma 3. □

5 Return to regression

Let W° be the full-data minimum-norm minimizer and $\rho_i = W^\circ x_i - y_i$. The normal equations imply $\sum_i \rho_i x_i^\top = 0$, and for every B,

$$L_D(W^\circ + B) = L_D^* + N^{-1} \sum_i \|B x_i\|^2.$$

Suppose $K = \sum_i x_i x_i^\top \succ 0$. Write $v_i = K^{-1/2} x_i$ and $\ell_i = \|v_i\|^2$. For $\ell_i > 0$, set

$$p_i = \ell_i/2, \qquad u_i = \sqrt{2/\ell_i}\, v_i, \qquad w_i = \sqrt{2/\ell_i}\, \rho_i.$$

These satisfy (1), with $E = \sum_{\ell_i > 0} \|\rho_i\|^2 \le N L_D^*$. For $\Theta = (W - W^\circ) K^{1/2}$, the excess loss is $\|\Theta\|_F^2 / N$.

Given frame weights q, choose original weights proportional to q_i/ℓ_i. The selected objective, in Θ coordinates, is a positive multiple of $\sum_i q_i \|\Theta u_i + w_i\|^2$. If $U(q) = \sum_i q_i u_i u_i^\top \succ 0$, its unique minimizer is $\Theta = -G(q)$, where

$$G(q) = \left(\sum_i q_i w_i u_i^\top \right) U(q)^{-1} = \sum_{i<j : D_{ij} > 0} \frac{q_i q_j D_{ij}}{\sum_{a<b} q_a q_b D_{ab}} G_{ij}, \qquad D_{ij} = \det(u_i, u_j)^2. \tag{9}$$

This follows by expanding the determinant and adjugate in dimension two: diagonal terms vanish and the i, j numerator term is $D_{ij} G_{ij}$.

The hulls defining $\mathcal{T}_3$ are closures of full-rank fits on the same support (cf. Zhang [3], Lemma E.1). For three independent pairs, positive prescribed coefficients β are attained by $q_i : q_j : q_k = D_{jk}/\beta_{jk}$:

$D_{ik}/\beta_{ik} : D_{ij}/\beta_{ij}$; boundary coefficients follow by limits. For an anchored segment with coefficients α_j, take

$$q_i = 1 - t, \qquad q_j = t\frac{\alpha_j/D_{ij}}{\sum_k \alpha_k/D_{ik}}, \qquad 0 < t < 1.$$

Star-pair coefficients dominate the leaf pair as $t \downarrow 0$, so $G(q) \to \sum_j \alpha_j G_{ij}$. This also covers parallel leaves; endpoints use an independent pair. Every approximating fit is full rank. Consequently

$$\inf_{|\operatorname{supp} c| \le 3} L_D(\widehat{W}_c) \le L_D^* + \mathcal{T}_3/N \le (11/6 - 10^{-20})L_D^*.$$

No invariance of minimum-norm tie breaking under whitening is needed.

If the feature span has dimension one, write $x_i = a_i e$ and $W^\circ = \beta e^\top$. On nonzero a_i, the planar vectors ρ_i/a_i are centered with probabilities $a_i^2/\sum_j a_j^2$. Represent zero by at most three of them and choose original weights proportional to the representation coefficients divided by a_i^2. The selected normal equation recovers β; minimum norm sets the action on $e^\perp$ to zero. Rank zero is immediate. If $E = 0$, a feature basis of size at most two recovers W° on the feature span, including the zero-loss case. These cases have ratio one. This proves Theorem 1. The lower bound $13/8$ is Zhang's Proposition 6.3 [3].

6 Formal verification

Source and Lean code are available at https://github.com/nZiben/weighted-selection-gap. The file `lean/Scalar.lean` verifies the two polynomial identities, finite scalar moment selection, the bound $5/6 - (11/60)(2r-1)^2$, sharpness of the scalar relaxation, the ratio bound $26/9$, and the exact numerical implications used above. The scalar selection theorems assume only nonnegativity, normalization and the two stated moments; they do not assume a selection bound. The final numerical contradiction is explicitly conditional on the geometric estimates proved in this paper. The planar selection and stability lemmas, support reduction, and regression bridge are not formalized; thus this is a partial formalization, not a Lean proof of Theorem 1. The checked source uses Lean 4.34.0 and a pinned Mathlib revision. The repository contains the compiler record and reproduction commands. The reported theorem axioms are only `propext`, `Classical.choice` and `Quot.sound`; no proof placeholders or additional axioms are used.

Use of AI

The initial proof was produced by GPT-6 Pro (GPT-6 Astra) in the ChatGPT user interface.

References

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